



The impact of low-carbon city policy on corporate cost stickiness[☆]

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ABSTRACT

We examine the impact of low-carbon city policy on corporate cost stickiness. Using a sample of Chinese A-share firms from 2008 to 2022, we document that cost stickiness increases for firms in a city with a low-carbon policy using a difference-in-differences research design. The findings are robust to an entropy balance method, alternative samples, placebo tests, after accounted for other environmental regulations, and after control for various time trend effects. The mechanism analysis suggests that easing financial constraints and engaging in green investments are the underlying logic behind the impact of low-carbon city policy on corporate cost stickiness. Additional analysis shows that the effect is more salient for firms in traditional manufacturing industries, which are state-owned, or belong to high technology industries. The economic analysis shows that while a firm's productivity generally decreases due to cost stickiness increase, the implementation of the low-carbon city policy offers significant benefits to offset the productivity decrease.

1. Introduction

Air pollution is a global issue. Countries attempt various policies to reach carbon-neutrality. As an emerging market, China has a more than fair-share of air pollution challenge. To mitigate the adverse impact of carbon emissions, China proposes to achieve carbon-neutrality by 2060.¹ As part of a policy to achieve the goal, China's National Development and Reform Commission designated cities in six provinces (all cities in these provinces), 80 cities, and one region (all cities in the region) to undergo a low-carbon policy in 2010, 2012, and 2017. We denote these cities as low-carbon policy cities. Accompanied the designation, cities implement a comprehensive policy to encourage sustainability. Several studies examine the impact of such a policy on corporate competitiveness (Duan & Xu, 2023; Zhu & Sun, 2022), innovation (Chen et al., 2021; Huang et al., 2021), employment (Cao et al., 2023; Wu et al., 2024), and green innovation (Pei & Wang, 2022; Wang et al., 2022) and report interesting results. The effect of such a policy on the detailed operation of firms are underexplored. Among the various operations, we focus on cost stickiness. Cost stickiness represents how a firm adjusts its cost downward in smaller magnitude when revenue decreases relative to adjust the cost upward when revenue increases in the same amount (Anderson et al., 2003). It is a core research question in management accounting. In the implementation of a low-carbon city policy, it unavoidably impacts funding, labor, capital, and technology. By examining the cost adjustment for firms in a low-carbon city

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¹ On September 22, 2020, the China Community Party Secretary Xi Jinping made the announcement during a United Nation briefing.

vis-à-vis not in a low-carbon city, we shed lights on the effect of the environmental policy on firm operation.

We hypothesize that a low-carbon city policy can ease corporate financial constraints due to the general increase in government funding or a lower tax from the policy. In addition, such a policy encourages corporate green innovation engagement. Then, the eased financial constraints drive management optimism so that managers lower their magnitude of downward cost adjustment. Similarly, green innovation projects are typically long-term. The long-term nature means high adjustment cost in the future when sales decrease. In both scenarios, cost stickiness increases. We test our hypothesis using the natural experiment of low-carbon city policy in China.

Using a sample of Chinese A-share firms from 2008 to 2022, we document that cost stickiness increases for firms in a city with a low-carbon policy using a difference-in-differences (DID) research design. The findings are robust to an entropy balance method, alternative samples, placebo tests, after accounted for other environmental regulations, and after control for various time trend effects. Mechanism analysis suggests that easing financial constraints and engaging in green investments are the underlying reasons behind the impact of low-carbon city policy on corporate cost stickiness. Additional analysis shows that the effect is more salient for firms in traditional manufacturing industries, which are state-owned, or belong to high technology industries. The economic analysis shows that while a firm's productivity generally decreases due to cost stickiness increase, the implementation of the low-carbon city policy offers significant benefits to offset the productivity decrease.

We contribute to the literature in several major ways. First, we advance the literature on the impact of a low-carbon city policy on corporate activities. Besides competitiveness, innovation, employment, and green innovation, we document such a policy increase corporate cost stickiness. Hence, we expand the effect of a low-carbon city on the economic and financial consequence from a management accounting perspective. Second, we complement the large body of literature on cost stickiness. These studies primarily focus on the internal contributing factors of cost stickiness (Chang et al., 2022; Chen et al., 2012; Dai et al., 2023). Few studies explore the effect of external factors, especially environmental-related ones, on cost stickiness. We show that a low-carbon city policy contributes to a higher cost stickiness.

Third, our findings provide lessons to other emerging markets in developing their own low-carbon city policies. Besides the typical economic and financial impact, firms need to manage their cost stickiness challenges when facing national environmental policy initiatives.

2. Background, literature review, and hypothesis development

2.1. Background

A low-carbon city is one that manages the behavior of firms, residents, and a government with the goal of low-carbon economy, low-carbon lifestyle, and low-carbon society. The World Wildlife Fund pointed out that the main purpose of establishing low-carbon cities is to reduce energy consumption and carbon dioxide emissions. In response to global climate change and the construction of low-carbon cities, China began implementing three batches of low-carbon city pilot projects beginning in 2010. The low-carbon city policy asks a city to develop a comprehensive low-carbon urban development plan, establish a greenhouse gas emission assessment system, and actively improve the innovation level and management ability of low-carbon development. At the same time, the low-carbon city policy also requires the city government to provide support for firms in industrial policies, fiscal and tax policies, and technology promotion policies to help accelerate low-carbon transformation. The Chinese low-carbon city policy applies to selected cities directly and all cities in selected provinces (in the first batch).

Specifically, the Chinese National Development and Reform Commission issued a notice on conducting pilot work in low-carbon provinces, regions, and cities on July 19, 2010. The first batch of provinces and cities include: Guangdong Province, Liaoning Province, Hubei Province, Shanxi Province, Yunnan Province, Tianjin City, Chongqing City, Shenzhen City, Xiamen City, Hangzhou City, Nanchang City, Guiyang City, and Baoding City. The second and third batches of cities were announced on November 26, 2012 and January 7, 2017. The low carbon city designations represent a new type environmental policy, which allows us to use a staggered difference-in-differences (DID) research design to examine the effect of a low-carbon city policy on a firm's cost stickiness.

2.2. Literature review

2.2.1. The effect of a low-carbon city policy on corporate activities

Several studies examine the effects of a low-carbon city policy at the city-level economy (Yang & Xu, 2023; Zhang et al., 2023) and the environment (Yang et al., 2023; Zeng et al., 2023).

An emerging body of literature studies the effect of a low-carbon policy on firms. Specifically, Zhu & Sun (2022) report that such a policy enhances corporate competitive positions in export. In contrast, Duan & Xu (2023) document that such a policy increases the operation costs of firms due to the compliance of the environmental initiatives and discourages corporate innovation, leading to less competitiveness in the marketplace.

Interestingly, Huang et al. (2021) leverage the low-carbon city policy to adopt a DID model to show that the policy promotes corporate R&D investment. Chen et al. (2021) find that the low-carbon policy enhances corporate total factor productivity. Qu et al. (2023) show that firms in low-carbon cities spearhead energy-related innovation and crowd-out other innovation. However, Cai et al. (2023) articulate that the low-carbon city policy increases corporate environment-compliance cost and hurts a firm's future expansion.

In terms of employment, Cao et al. (2023) find that firms in a low-carbon city have lower agency cost and more green innovation. These firms hire more high-quality employees and the process enhances labor force upgrade. Similarly, Wu et al. (2024) show that low-carbon cities streamline their government expenditure to restrain state-owned firms' excessive capital projects. The low-carbon

initiatives increase hiring labor. Consequently, the capital-labor ratio at the city level lowers.

With respect to green innovation, Wang et al. (2022) document that, as expected, such a policy promotes green innovation. Similarly, Pei & Wang (2022) report that corporate green patent applications increase in firms located in low-carbon cities. Zhuo (2023) examines corporate sustainability and find a low-carbon city promote sustainability.

Overall, while several studies examine the impact of a low-carbon city policy on corporate activities. The evidence focuses on innovation, employment, and green innovation with some studies offer mixed evidence. While the literature generally suggests that a low-carbon city policy raises the compliance cost of firms, we do not know how stickiness cost changes with such policy.

2.2.2. Cost stickiness

Anderson et al. (2003) document that the magnitude of cost increase during revenue increase for a firm is less than the magnitude when there is the same extent of revenue decrease in a sample of US firms. Cost adjustment downward and upward in response to revenue change is asymmetric. It is a phenomenon of cost stickiness.

Banker & Byzlov (2014) summarize three reasons for cost stickiness: cost adjustment challenge, management optimism, and agency problems. From the cost adjustment perspective, management considers friction in downward cost adjustments, especially with the cost of labor. For instance, when revenue decreases, a firm needs to pay severance compensation to lay off employees. In contrast, when revenue increases, a firm can simply hire more employees without additional cost. When the adjustment cost (severance compensation) is high, firms reduce their downward cost adjustment. Then, cost stickiness occurs. Banker et al. (2013) use labor protection law to proxy for adjustment cost and report evidence of cost stickiness in the US. Similarly, Holzhacker et al. (2015) examine medical supply firms in Germany and find that price control leads to cost stickiness. Belina et al. (2019), Chang et al. (2022) and Dai et al. (2023) use different setting of adjustment cost due to medical insurance claim, labor union, and credit default swap usage to reveal evidence of cost stickiness.

In terms of management optimism, Banker et al. (2014) suggest that when revenue increases, optimistic managers are willing to raise cost to support revenue. When revenue decrease, optimistic managers believe that the revenue can bounce back. Thus, these managers are slow to downward adjust cost. Chen et al. (2019) report that when a firm reveals management optimism in its management disclosure and analysis section of an annual report, its cost stickiness is large.

With respect to agency problem, it considers that management is opportunistic. A lower cost structure suggests a lesser pool of resource available for private interests of management. Hence, when revenue decrease, cost does not decrease in the same magnitude as cost increases when revenue increases. Chen et al. (2012) document that cost stickiness is positively correlated with free cashflows. Kama & Weiss (2013) show that management downward adjusts slack resource to lower cost stickiness to meet revenue target, suggesting the motivation behind cost stickiness is management's desire to meet their performance target. Xu et al. (2023) report higher cost stickiness for non-business group privately-owned Chinese firms than those of firms belong to business groups. Because business group firms typically have lower agency conflicts, the results in Xu et al. (2023) suggest agency problem play a role in cost stickiness.

The cost stickiness literature primarily emphasizes the role a firm's internal factor but underexplores how a low-carbon city policy may contribute to the cost stickiness of a firm.

2.2.3. Hypothesis development

We propose two major conceptual frameworks to develop a testable hypothesis. First, a low-carbon city policy suggests additional funding injected in the cities where firms are located. The funding opportunities include low taxes, government subsidies, or green loans from banks. For firms to engage in environmental investment, they have more cashflows. Hence, firms can move the intended funds for environmental projects to finance other non-financial projects. All else the same, management optimism increases. When revenue increases, management raises their cost to support revenue. When revenue decreases, management's optimism drives them to tolerate operation slack. Consequently, cost stickiness increases after the implementation of low-carbon city policy.

Second, the low-carbon city policy suggests that firms are under close watch in the environmental performance. We expect that firms engage in more green innovation to limit their pollution at the frontend. According to Porter hypothesis (Porter & Linde, 1995), a firm can benefit from green innovation more than the cost to conduct green innovation. The low-carbon city officials naturally encourage firms in their jurisdictions to engage in green innovation. Green innovation, similar to other innovation projects, requires substantial corporate investments. Once green projects are underway, it is costly or impossible to back out. It is a cost adjustment phenomenon and results in cost stickiness increases. In sum, we have the following hypothesis:

H1. Cost stickiness increases when a firm is in a low-carbon city

We argue that the low-carbon city pilot policy, as an important environmental regulation policy, will strengthen the cost stickiness of firms by alleviating financial constraints and improving green innovation levels. On the one hand, the governments of low-carbon cities have provided significant financial support for firms to accelerate their low-carbon transformation, which helps alleviate financial constraints. Specifically, the low-carbon city policy can help alleviate the financial constraints of a firm due to a pilot city's low-carbon related subsidies or improved financing from lenders as a result of enhanced corporate environmental performance. The alleviation of financial constraints drives better financial flexibility to the firm. Then, it enhances the optimistic expectations of managers, and ultimately lead to an increase in cost stickiness. Moreover, based on the optimistic expectations of managers, the alleviation of financial constraints in firms provides a certain degree of risk buffering, which will further increase managers' preference for investing in risky projects and strengthen their optimistic expectations. When performance declines, managers with an optimistic mindset are more willing to expect future performance improvement, thereby retaining existing idle resources and increasing cost

stickiness.

On the other hand, the low-carbon city policy can help motivate firms to engage in green innovation. The low-carbon city policy encourages firms in pilot areas to introduce green technologies and develop green products based on their actual situation, and also encourages green technology learning and cooperation among enterprises, which helps them improve their green innovation level. At the same time, the pilot area governments also provide technical information, talent support, government subsidies, and tax reductions for firms to carry out green innovation, which helps alleviate the funding problem for firms to carry out green innovation and helps them improve their green innovation. Considering the high risk, long cycle, and large investment characteristics of green innovation, managers will not easily reduce human and material resources related to green research and development after adjusting green R&D investment upwards, resulting in increased cost asymmetry and increased cost stickiness.

Based on the analysis above, we propose a hypothesis:

H2. The low-carbon city policy increases corporate cost stickiness by alleviating financial constraints and promoting green innovation.

3. Research design

3.1. Sample

We begin with all A-share manufacturing firms on the Shanghai and Shenzhen Stock Exchanges from 2008 to 2022. Manufacturing firms generally have large amount of fixed assets. They naturally produce pollutants and they employ large number of workers. Moreover, manufacturing industries are competitive and hence, they are subject to less monopolistic power (Banker et al., 2014). Thus, these firms respond to environmental regulatory changes to make cost stickiness analysis meaningful. We began our analysis in 2008 because new accounting reporting requirements began in 2008 and low carbon city designations began in 2010 in China. The accounting and financial data are obtained from the China Stock Markets and Accounting Research database, green patent information is from the Chinese Research Data Services Platform and ESG is from the WIND database. The initial sample size has 19,433 firm-years.

Then, we delete: (1) 228 financially distressed firms (ST and ST* designated firms), (2) 74 firms with the ratio of total liabilities to total assets exceed 1, (3) 58 firms with sales, general, and administrative expense exceeds revenue, (4) 5072 firms without two consecutive years of revenue or sales, general, and administrative expense, and (5) 1124 firms with missing accounting or financial data. The final sample has 2203 firms and 12,877 firm-years. We winsorize all continuous variables at the 1% level. Table 1 presents the sample selection process.

3.2. Model

We follow Anderson et al. (2003), Chen et al. (2012), and Xu et al. (2023) to use the cost stickiness model of Eq. (1).

$$\begin{aligned} \Delta LNSGA_{i,t} = & \alpha_0 + \alpha_1 \Delta LNREV_{i,t} + \alpha_2 \Delta LNREV_{i,t} * D_{i,t} \\ & + \alpha_3 \Delta LNREV_{i,t} * D_{i,t} * LCARBON_{i,t} + \alpha_4 LCARBON_{i,t} \\ & + \alpha_5 \Sigma \Delta LNREV_{i,t} * D_{i,t} * ECONVAR_{i,t} + ECONVAR_{i,t} \\ & + CONTROL + FIRM + YEAR + \varepsilon_{i,t} \end{aligned} \quad (1)$$

Where $\Delta LNSGA$ is the change in natural logarithm of sales, general, and administrative expense to capture a firm's change in operating cost, $\Delta LNREV$ is the change in natural logarithm of revenue a firm, D is a (1, 0) indicator variable with a value of 1 if a firm experiences a decrease in revenue at t relative to $t-1$ and 0 otherwise, and $LCARBON$ is an (1, 0) indicator variable with a value of 1 if a firm is located in a low-carbon policy city at t and 0 otherwise.

$ECONVAR$ represents a set of economic control variables include: (1) an (1, 0) indicator variable, $DECREASE$, with a value of 1 if a firm has two consecutive years negative revenue growth and 0 otherwise, (2) GDP growth rate in China ($GDPGROWTH$), (3) asset intensity of a firm ($AINENSITY$), which is the ratio of total revenue to total assets, and (4) employee intensity ($EINTENSITY$), which is

Table 1
Sample selection.

	Deleted	N	No. of firms
Beginning sample:			
A-share firms in the manufacturing industries 2008–2022		19,433	2833
Deleted			
ST and ST* firms	228		
Firms with the ratio of total liabilities to total assets exceeds 1	74		
Firms with sales, general, and administrative expense exceeds revenue	58		
Firms without two consecutive years of revenue or sales, general, and administrative expense	5072		
Firms with missing accounting or financial data	1124		
Remaining		12,877	2203

the ratio of total revenue to the total number of employees (in millions RMB). We follow [Anderson et al. \(2003\)](#) to account for the economic impact of these variables on a firm's operating cost.

CONTROL is a set of other firm level control variables include: (1) management shareholding percentage (*MSH*), (2) firm age (*AGE*), (3) shareholder balance (*SHB*) which is the ratio of the second to fifth largest shareholders' ownership to those of the largest shareholder, (4) firm size (*SIZE*), (5) return on asset (*ROA*), (6) financial leverage (*LEV*), and (7) state ownership of the firm (*SOE*). Detailed information of all variables are presented in [Appendix A](#). We also include firm and year fixed effects in Eq. (1).

The cost stickiness is captured by the coefficient of $\Delta LNREV \cdot D$. When there is an increase in revenue ($D = 0$), α_1 captures the percentage increase in cost when revenue increases by 1%. When there is a decrease in revenue ($D = 1$), $\alpha_1 + \alpha_2$ captures the percentage decrease in cost when revenue decreases by 1%. If $\alpha_2 < 0$, then $(\alpha_1 + \alpha_2) < \alpha_1$, which implies a cost stickiness. In similar logic, the coefficient of $\Delta LNREV \cdot D \cdot LCARBON$ (α_3) captures the effect of a low carbon city designation on cost stickiness. If [H1](#) is valid, $\alpha_3 < 0$.

4. Results and discussions

4.1. Summary statistics

We present the summary statistics of the sample in [Table 2](#). The mean and standard deviation of $\Delta LNREV$ are 0.1324 and 0.2566. Hence, on average, a firm has approximately 13.24% of revenue growth but the variation is large. In terms of $\Delta LNPGA$, the mean and standard deviation are 0.0961 and 0.2361. Hence, the cost of a sample firm, on average, increases 9.61% with large variation. The mean of D is 0.2584, suggesting that 25.84% of the sample firm exhibits a decrease in revenue in a given year. The mean of $LCARBON$ is 0.6070, indicating that approximately 60.7% of the sample firm-years are in a low-carbon policy city.

4.2. Baseline results

We present the baseline results of Eq. (1) in [Table 3](#). Besides the full model, we include various versions of Eq. (1). Across four columns, the coefficients of $\Delta LNREV \cdot D$ are consistently negative and significant at the 1% level. Hence, the firms in our sample, on average, exhibit cost stickiness. The results echo those in the literature ([Anderson et al., 2003](#); [Xu et al., 2023](#)). For cost stickiness related to low carbon city designation, we notice the coefficients of $\Delta LNREV \cdot D \cdot LCARBON$ in columns (2) to (4) are negatively significant at the 1% level, suggesting that when a firm is in a low carbon city, its cost stickiness increases beyond the normal cost stickiness level. Using column (4), the magnitude of the coefficient of $\Delta LNREV \cdot D \cdot LCARBON$ is -0.1066 . Hence, when a firm is located in a low carbon city, its decrease in cost is 0.1066 less than a firm not in a low carbon city, which is approximately 111% of an average firm ($0.1066/0.0961$). When compared to the magnitude of the coefficient of $\Delta LNREV \cdot D$ (which is -0.1599), the additional "cost stickiness" due to low carbon impact is approximately 67% ($0.1066/0.1599$). Thus, when a firm is in a low carbon city, the cost stickiness becomes more severe. The results support [H1](#).

Table 2
Summary statistics

[Table 2](#) presents summary statistics of the sample. Variable definitions are presented in [Appendix A](#).

Variable	N	Mean	Median	Std Dev.	Min	Max
$\Delta LNPGA$	12877	0.0961	0.0982	0.2361	-0.5836	0.8286
$SGA(\text{\textyen millions})$	12877	451.2583	186.216	1442.3392	6.4076	44142.5938
$\Delta LNREV$	12877	0.1324	0.1205	0.2566	-0.587	1.0285
$REV(\text{\textyen millions})$	12877	4360.9806	1422.6715	16769.3269	19.0456	5.12E+05
D	12877	0.2584	0	0.4378	0	1
$LCARBON$	12877	0.607	1	0.4884	0	1
$DECREASE$	12877	0.0866	0	0.2812	0	1
$GDPGROWTH$	12877	6.1785	6.734	2.0921	2.244	10.606
$AINENSITY$	12877	2.2129	1.8988	1.2513	0.548	7.7199
$EINTENSITY$	12877	1.4	1.2184	0.8766	0.1721	4.6754
MSH	12877	21.5976	16.2422	20.9266	0	70.2062
AGE	12877	1.7759	1.7918	0.5202	0.6931	2.7726
SHB	12877	0.8414	0.6766	0.6216	0.0565	2.9421
$SIZE$	12877	21.8234	21.7189	0.9907	19.9507	25.1311
ROA	12877	0.0406	0.0418	0.0632	-0.2491	0.1999
LEV	12877	0.3645	0.3553	0.1742	0.0563	0.7744
SOE	12877	0.1172	0	0.3217	0	1
$ABSSA$	12877	3.1703	3.1948	0.0852	2.2577	3.2727
$GPATENT$	12862	4.3932	0	26.4484	0	1154
$INTANGIBLE (\text{\textyen millions})$	12874	298.2259	103.9748	1178.3765	0.00002	5362.9764
TFP_{OP}	12374	7.5002	7.4340	0.7640	4.5798	11.3382
$ABSSTICKY$	11862	0.7442	0.4264	0.8992	0.0001	10.1653

Table 3

Baseline results

Table 3 reports the baseline results for the impact of low-carbon city policy on a firm's cost stickiness. T-statistics are in parentheses. Appendix A presents the definitions of all variables. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dep. Var: $\Delta LNSGA$	Base cost stickiness model	Base cost stickiness model with low carbon city	Included firm level control variables	Included firm level and economy control variables
	(1)	(2)	(3)	(4)
$\Delta LNREV$	0.5293*** (40.3575)	0.5293*** (40.3869)	0.5295*** (38.8383)	0.5307*** (37.2642)
$\Delta LNREV \cdot D$	-0.3495*** (-13.3591)	-0.2778*** (-8.1580)	-0.2449*** (-7.2569)	-0.1599** (-2.2882)
$\Delta LNREV \cdot D \cdot LCARBON$		-0.1158*** (-3.6060)	-0.1083*** (-3.6080)	-0.1066*** (-4.1649)
$LCARBON$		-0.0076 (-0.5079)	0.0022 (0.1434)	0.0000 (0.0021)
MSH			0.0006* (1.8026)	0.0005 (1.6359)
AGE			-0.0580*** (-3.3501)	-0.0572*** (-3.3731)
SHB			0.0073 (1.3259)	0.0095* (1.7719)
$SIZE$			0.0646*** (8.8603)	0.0660*** (7.8686)
ROA			-0.3824*** (-8.8418)	-0.3649*** (-8.7827)
LEV			-0.0266 (-1.1615)	-0.0246 (-0.9724)
SOE			-0.0165 (-0.8289)	-0.0163 (-0.8384)
$\Delta LNREV \cdot D \cdot DECREASE$				0.1091** (2.3254)
$\Delta LNREV \cdot D \cdot GDPGROWTH$				-0.0036 (-0.4281)
$\Delta LNREV \cdot D \cdot AINTENSITY$				-0.0359** (-2.5985)
$\Delta LNREV \cdot D \cdot EINTENSITY$				0.0018 (0.0716)
$DECREASE$				-0.0213** (-2.1409)
$AINTENSITY$				-0.0035 (-0.8251)
$EINTENSITY$				0.0234*** (5.4153)
Constant	0.0115*** (4.6424)	0.0161 (1.6734)	-1.2855*** (-8.9280)	-1.3433*** (-8.4439)
N	12, 877	12, 877	12, 877	12, 877
FIRM	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Adj. R ²	0.4166	0.4171	0.4301	0.4339

4.3. Robustness checks

4.3.1. Parallel trend analysis

A DID research design in Eq. (1) requires a parallel trend assumption. To examine the validity of the assumption, we follow Huang et al. (2019) to use Eq. (2) below:

$$\begin{aligned}
 \Delta LNSGA_{i,t} = & \beta_0 + \beta_1 \Delta LNREV_{i,t} + \beta_2 \Delta LNREV_{i,t} \cdot D_{i,t} \\
 & + \beta_3 \Delta LNREV_{i,t} \cdot D_{i,t} \cdot BEFORE4_{i,t} + \beta_4 \Delta LNREV_{i,t} \cdot D_{i,t} \cdot BEFORE3_{i,t} \\
 & + \beta_5 \Delta LNREV_{i,t} \cdot D_{i,t} \cdot BEFORE2_{i,t} + \beta_6 \Delta LNREV_{i,t} \cdot D_{i,t} \cdot BEFORE1_{i,t} \\
 & + \beta_7 \Delta LNREV_{i,t} \cdot D_{i,t} \cdot AFTER1_{i,t} \\
 & + \beta_8 \Delta LNREV_{i,t} \cdot D_{i,t} \cdot AFTER2_{i,t} + \beta_9 \Delta LNREV_{i,t} \cdot D_{i,t} \cdot AFTER3_{i,t} \\
 & + \beta_{10} \Delta LNREV_{i,t} \cdot D_{i,t} \cdot AFTER4_{i,t} \\
 & + \beta_{11} BEFORE4_{i,t} + \beta_{12} BEFORE3_{i,t} + \beta_{13} BEFORE2_{i,t} + \beta_{14} BEFORE1_{i,t}
 \end{aligned}$$

$$\begin{aligned}
& + \beta_{15} \text{AFTER1}_{i,t} + \beta_{16} \text{AFTER2}_{i,t} + \beta_{17} \text{AFTER3}_{i,t} + \beta_{18} \text{AFTER4}_{i,t} \\
& + \beta_{19} \Delta \text{LNREV}_{i,t} * D_{i,t} * \text{ECONVAR}_{i,t} + \text{ECONVAR} \\
& + \text{CONTROL} + \text{FIRM} + \text{YEAR} + \varepsilon_{i,t}
\end{aligned} \tag{2}$$

Where *BEFORE4*, *BEFORE3*, *BEFORE2*, and *BEFORE1* are dummy variables with values of 1 if they are in $t-4$, $t-3$, $t-2$, and $t-1$ years, respectively, before a city is designated as a low carbon city at t and 0 otherwise. Similarly, we define *AFTER1*, *AFTER2*, *AFTER3*, and *AFTER4* as dummy variables with values of 1 if the firm is in 1, 2, 3, and 4 years, respectively, after their located city received low carbon city designation after t and 0 otherwise. We do not put current year in Eq. (2) because we use it as the baseline year for comparison.

We present the findings for Eq. (2) in Panel A of Table 4. The coefficients of $\Delta \text{LNREV} * D * \text{BEFORE4}$, $\Delta \text{LNREV} * D * \text{BEFORE3}$, $\Delta \text{LNREV} * D * \text{BEFORE2}$, and $\Delta \text{LNREV} * D * \text{BEFORE1}$ are insignificant, suggesting that before low carbon city designation, the impact of the designation on cost stickiness is minimal. Hence, the parallel trend assumption of Eq. (1) is valid. Among the other variables, the coefficient of $\Delta \text{LNREV} * D * \text{AFTER4}$ is negatively significant at the 5% level. That is, after the low carbon city designation at t , a firm in such a city begins to exhibit cost stickiness four years later.

4.3.2. Placebo test

To address the concern that the results in Table 3 are due to random factors, we conduct a placebo test. We follow Gao et al. (2020) to randomly replicate the low carbon city date 1000 times to estimate Eq. (1) and plot the coefficients of $\Delta \text{LNREV} * D * \text{LCARBON}$ in Fig. 1. The distribution of the coefficients concentrates around 0, suggesting that our findings in Table 3 is not due to random factors. The coefficient of $\Delta \text{LNREV} * D * \text{LCARBON}$ (−0.1066) in column (4) of Table 3 is on the left-hand side of the distribution, which consistent with the placebo test results.

4.3.3. Entropy balancing result

We follow Hainmueller (2012) to conduct an entropy balancing test on the baseline results in Eq. (1). Such a method is superior to a propensity score matching (Ferri & Li, 2020). Entropy balancing applies a weighing scheme to match treatment and control firms. The steps are: (1) we use all control variables in Appendix A as covariates that will affect policy evaluation (here we select all control variables in Appendix A) and find a set of weights to ensure that all covariates in the treatment group and control group samples are the same in terms of mean, variance, and skewness; and (2) based on the obtained weights, we assign higher weights to control groups similar to the treatment group, and then use weighted regression to estimate the regression model. Appendix B presents the matching results after the entropy balancing. After the entropy balancing, the mean, variance, and skewness of treatment and control firms are almost identical. Then, we reexamine Eq. (1) using the matched samples. The results in column (1) of Table 4 Panel B show that the coefficient of $\Delta \text{LNREV} * D * \text{LCARBON}$ continues to be negatively significant at the 1% level. The magnitude of $\Delta \text{LNREV} * D * \text{LCARBON}$ is −0.0881, which is comparable to the −0.1066 in Table 3.

4.3.4. Controlling the impact of ESG

The corporate ESG can alleviate financial constraints (Bai et al., 2022), promote green innovation (Li et al., 2023) and increase cost stickiness (Lin & Fang, 2024). Therefore, to avoid the influence of ESG on our regression results, we add corporate ESG performance to Eq. (1) as a robust test. We use quarterly averages of Huazheng ESG rating² data to measure corporate ESG performance. In Column (2) of Panel B in Table 4, the coefficient of $\Delta \text{LNREV} * D * \text{LCARBON}$ is significantly negative at the 1% level, indicating that our main regression results remain robust after controlling for ESG performance.

4.3.5. Alternative samples

For robustness, we use several alternative samples to reexamine Eq. (1). First, some cities are designated as low carbon cities while their provinces are also designated as low carbon provinces in earlier years. To make the sample clean, we exclude these firms in the designated cities. The results in column (1) of Table 5 show that the coefficient of $\Delta \text{LNREV} * D * \text{LCARBON}$ remains negatively significant at the 1% level.

Second, instead of deleting firms in the low-carbon policy in column (1) of Table 5, we delete firms in the designated provinces in the first batch of low-carbon provinces and keep firms in the designated cities. The results in column (2) of Table 5 are qualitatively similar to those in column (1).

Third, autonomous cities (Beijing, Shanghai, Tianjin, and Chongqing) are enormously large. For robustness, we exclude firms in these cities. The results in column (3) of Table 5 show that the coefficient of $\Delta \text{LNREV} * D * \text{LCARBON}$ continues to be negatively significant at the 1% level.

Fourth, in order to increase the generalizability of our research results, we also use all industry samples for regression. The fourth column of Table 5 shows the regression results of the entire industry sample. The coefficient of $\Delta \text{LNREV} * D * \text{LCARBON}$ is significantly negative at the 10% level, indicating that even if our research findings are extended to all industries, the baseline results remain intact.

Fifth, our sample spans from 2008 to 2022, a period that includes significant global and domestic economic events (e.g., the 2008 financial crisis, COVID-19 pandemic). Therefore, we select 2010–2019 as the sample period for the robustness test to exclude the impact of the 2008 financial crisis and the COVID-19 pandemic. The fifth column of Table 5 shows the results of the regression using

² We assign values from 1 to 9 according to the Huazheng ESG rating data, the higher values indicating higher ratings.

Table 4

Parallel trend analysis and other robustness checks

Table 4 shows the results of parallel trend analysis and other robustness checks. Panel A reports the parallel trend analysis for the impact of low-carbon city policy on a firm's cost stickiness. *BEFORE4*, *BEFORE3*, *BEFORE2*, and *BEFORE1* are dummy variables with values of 1 if they are in $t-4$, $t-3$, $t-2$, and $t-1$ years, respectively, before a city is designated as a low carbon city at t and 0 otherwise. Similarly, we define *AFTER1*, *AFTER2*, *AFTER3*, and *AFTER4* as dummy variables with values of 1 if the firm is in 1, 2, 3, and 4 years, respectively, after their located city received low carbon city designation after t and 0 otherwise. In Panel B, we report the results of various robustness checks. Column (1) presents the findings using entropy balancing. Columns (2) controls ESG performance. We use quarterly averages of Huazheng ESG rating data to measure corporate ESG performance. For brevity, we do not include the coefficients of control variables. T-statistics are in parentheses. Appendix A presents the definitions of all variables. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Parallel trend analysis		
Dep. Var: $\Delta LNSGA$	Parallel trend analysis	
	(1)	
$\Delta LNREV$	0.5308***	
	(36.5658)	
$\Delta LNREV * D$	-0.2413***	
	(-3.9056)	
$\Delta LNREV * D * BEFORE4$	-0.1304	
	(-0.8840)	
$\Delta LNREV * D * BEFORE3$	0.1035	
	(0.5315)	
$\Delta LNREV * D * BEFORE2$	-0.1429	
	(-0.8538)	
$\Delta LNREV * D * BEFORE1$	0.1059	
	(1.1439)	
$\Delta LNREV * D * AFTER1$	-0.1486	
	(-1.5393)	
$\Delta LNREV * D * AFTER2$	0.0679	
	(0.4526)	
$\Delta LNREV * D * AFTER3$	-0.0824	
	(-0.7900)	
$\Delta LNREV * D * AFTER4$	-0.1841**	
	(-2.1114)	
<i>BEFORE4</i>	0.0132	
	(0.6892)	
<i>BEFORE3</i>	0.0089	
	(0.4482)	
<i>BEFORE2</i>	0.0285	
	(1.3396)	
<i>BEFORE1</i>	0.0080	
	(0.6014)	
<i>AFTER1</i>	-0.0024	
	(-0.1353)	
<i>AFTER2</i>	-0.0054	
	(-0.9471)	
<i>AFTER3</i>	0.0141	
	(1.1118)	
<i>AFTER4</i>	0.0040	
	(0.3817)	
Other var.	Yes	
Constant	-1.3313***	
	(-8.4539)	
N	12, 877	
FIRM	Yes	
YEAR	Yes	
Adj. R ²	0.4337	
Panel B: Other robustness checks		
Dep. Var: $\Delta LNSGA$	Entropy balancing	Including ESG
	(1)	(2)
$\Delta LNREV$	0.5369***	0.5344***
	(27.5965)	(37.6446)
$\Delta LNREV * D$	-0.1869**	-0.1521**
	(-2.2088)	(-2.0643)
$\Delta LNREV * D * LCARBON$	-0.0881***	-0.1003***
	(-3.3428)	(-3.2339)
<i>LCARBON</i>	0.0083	-0.0010
	(0.5655)	(-0.0716)

(continued on next page)

Table 4 (continued)

Panel B: Other robustness checks		
Dep. Var: $\Delta LNSGA$	Entropy balancing	Including ESG
	(1)	(2)
ESG		0.0138*** (4.9562)
Other var.	Yes	Yes
Constant	−1.3664*** (−6.4834)	−1.3005*** (−8.1527)
N	12, 877	12,527
FIRM	Yes	Yes
YEAR	Yes	Yes
Adj. R ²	0.4428	0.4376

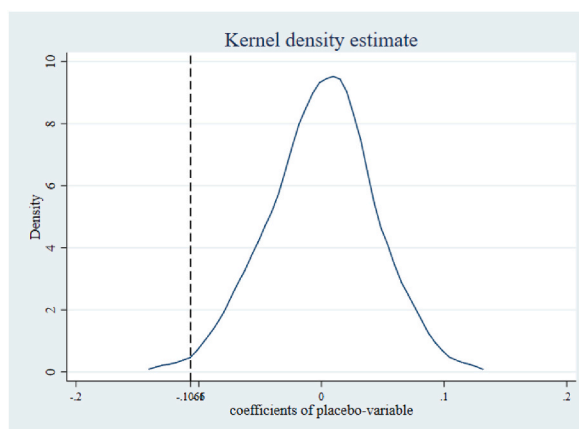


Fig. 1. The distribution of the coefficients of cost stickiness after 1000 times of random low carbon city designations.

Table 5

Alternative samples

Table 5 reports the results for alternative samples. Columns (1) to (3) delete samples for firms in low carbon cities that was previously designated in low carbon provinces, firms in low-carbon policy provinces, and firms in autonomous cities, respectively. Column (4) uses industry wide samples. Column (5) uses 2010–2019 as the sample period. Column (6) includes the financially distressed firms and those with extreme financial ratios. For brevity, we do not include the coefficients of control variables. T-statistics are in parentheses. Appendix A presents the definitions of all variables. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dep. Var: $\Delta LNSGA$	Delete firms in low carbon cities that was previously designated in low carbon provinces	Delete firms in low-carbon policy provinces	Delete firms in autonomous cities	Industry wide samples	Using 2010–2019 as the sample period	Including the financially distressed firms and those with extreme financial ratios
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta LNREV$	0.5309*** (48.4032)	0.5411*** (43.0159)	0.5288*** (46.0076)	0.5186*** (58.3347)	0.5955*** (45.7611)	0.5134*** (48.7277)
$\Delta LNREV^*D$	−0.1586** (−2.3483)	−0.2051*** (−2.6725)	−0.1348* (−1.9119)	−0.1794*** (−3.4879)	−0.0030 (−0.0122)	−0.1228** (−1.9642)
$\Delta LNREV^*D^*LCARBON$	−0.1068*** (−2.9136)	−0.0845** (−2.0272)	−0.1220*** (−3.2258)	−0.0547* (−1.8888)	−0.1066** (−2.2526)	−0.1168*** (−3.4234)
$LCARBON$	−0.0046 (−0.4356)	−0.0007 (−0.0611)	−0.0024 (−0.2169)	−0.0013 (−0.1579)	−0.0100 (−0.7909)	0.0039 (0.3688)
Other var.	Yes	Yes	Yes	Yes	Yes	Yes
Constant	−1.3961*** (−11.1377)	−1.3656*** (−9.5805)	−1.3721*** (−10.3542)	−1.4827*** (−15.0780)	−1.7486*** (−10.4602)	−1.6046*** (−13.5891)
N	12,059	9529	11,208	17,748	7993	13,088
FIRM	Yes	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.4363	0.4319	0.4294	0.4159	0.4793	0.4243

the sample period from 2008 to 2019. The regression results show that the coefficient of $\Delta LNREV * D * LCARBON$ is significantly negative at the 5% level, indicating that the main regression results are still robust after excluding the interference of the 2008 financial crisis and the COVID-19 epidemic.

Sixth, our sample excludes financially distressed firms and those with extreme financial ratios might introduce survivorship bias. Hence, we retain the financially distressed firms and those with extreme financial ratios in the sample and include them as robustness tests. Column (6) of Table 5 presents the regression results for retaining financially distressed firms and those with extreme financial ratios. The regression results show that the coefficient of $\Delta LNREV * D * LCARBON$ is negatively significant at the 1% level, indicating that the main regression results of this study are still robust when we consider the survival bias.

4.3.6. Consider other environmental regulations

During the sample period, there were four other major environmental regulations. First, the Chinese government announced a list of cities that undergo air pollution watch in 2012. We define *AIR* as a dummy variable with a value of 1 if firms locate in cities that undergo air pollution and the sample year is 2012 or later, and 0 otherwise.

Second, the Chinese government put a limit of air pollutants upon a list of cities in 2013. We define *LIMIT* as a dummy variable with a value of 1 if firms belong to the lists and the sample year is 2013 or later, and 0 otherwise.

Third, the “Green Credit Guidelines” policy formulated by the China Banking Regulatory Commission in 2012 provides financial support for firms to develop green innovation. Therefore, the green credit policy may have a significant impact on corporate financial constraints and green innovation (Li et al., 2018), which in turn may affect our regression results. To eliminate the interference of this policy, we incorporate the green credit policy *GCREDIT* into Eq. (1). We define *GCREDIT* as a dummy variable with a value of 1 if firms belong to a heavy polluting industry³ and the sample year is 2012 or later, and 0 otherwise.

Fourth, in 2011, the National Development and Reform Commission issued a notice on the pilot work of carbon emission trading, approving seven provinces and cities including Beijing, Shanghai, Tianjin, Chongqing, Hubei, Guangdong, and Shenzhen to carry out carbon trading pilot work. The carbon trading pilot work also occurred during our sample period, which may have an impact on the main regression results. We use *CTRADE* to measure carbon trading schemes. We define *CTRADE* as a dummy variable with a value of 1 if firms locate in a carbon trading pilot city and the sample year is 2011 or later, and 0 otherwise.

We present the findings in columns (1) and (4) of Table 6. The coefficients of $\Delta LNREV * D * LCARBON$ remain negatively significant at the 1% level. Overall, the results in Table 6 suggest that our baseline findings are robust.

4.3.7. Account for various time trend effects

In addition to year and firm fixed effects, we control various additional time trend effect with respect to city (*CITY*), industry (*INDUSTRY*) special economic zone (*ZONE*), provincial capital (*CAPITAL*), eastern region (*EAST*), and acid rain-sulfur dioxide watch cities (*WATCH*). The results in Table 7 show that the baseline findings remain qualitatively the same.

4.4. Transmission mechanism

We argue that alleviating a firm’s financial constraint and promoting green innovation are the underlying logic for the impact of low carbon city policy on cost stickiness. To examine the validity of the logic, we use a mediation effect model to further test our mechanism. Following Baron & Kenny (1986), we build models (3) and (4).

$$MED_{i,t} = \gamma_0 + \gamma_1 LCARBON_{i,t} + CONTROL + FIRM + YEAR + \rho_{i,t} \quad (3)$$

$$\begin{aligned} \Delta LNPGA_{i,t} = & \delta_0 + \delta_1 \Delta LNREV_{i,t} + \delta_2 \Delta LNREV_{i,t} * D_{i,t} \\ & + \delta_3 \Delta LNREV_{i,t} * D_{i,t} * LCARBON_{i,t} + \delta_4 LCARBON_{i,t} \\ & + \delta_5 \Delta LNREV_{i,t} * D_{i,t} * MED_{i,t} + \delta_6 MED_{i,t} \\ & + \delta_7 \Delta LNREV_{i,t} * D_{i,t} * ECONVAR_{i,t} + \delta_8 ECONVAR_{i,t} \\ & + CONTROL + FIRM + YEAR + \zeta_{i,t} \end{aligned} \quad (4)$$

Where *MED* represents intermediary variables, namely financial constraints and green innovation. In Eq. (3), we regard the mediator variable *MED* as the dependent variable and take the low-carbon city *LCARBON* as the independent variable. In Eq. (4), we add mediator variables to Eq. (1). The other variables and model settings are the same as Eq. (1).

Following Hadlock & Pierce (2010), we use the absolute value of SA to measure financial constraints. $ABSSA = |-0.737 * SA|$

³ According to the 2012 Industry Classification Standards of the China Securities Regulatory Commission and the Industry Classification Catalogue for Environmental Protection Inspection of Listed Companies, heavy polluting industries mainly include coal mining and washing industry, oil and gas mining industry, black metal mining and selection industry, non-ferrous metal mining and selection industry, non-metallic mineral mining and selection industry, alcohol, beverage and refined tea manufacturing industry, textile industry, textile and clothing industry, leather, fur, feather and its products and footwear industry, paper and paper products industry, petroleum processing, coking and nuclear fuel processing industry, chemical raw materials and chemical products manufacturing industry, pharmaceutical manufacturing industry, chemical fiber manufacturing industry, rubber and plastic products industry, non-ferrous metal smelting and rolling processing industry.

Table 6

Consider other environmental regulations

Table 6 reports the results of considering other environmental regulations. Columns (1) to (4) account for air pollution watch, air pollutant limit, green credit and carbon trading pilot work, respectively. For brevity, we do not include the coefficients of control variables. T-statistics are in parentheses. Appendix A presents the definitions of all variables. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dep. Var: $\Delta LNSGA$	Account for air pollution watch	Accounting for air pollutant limit	Account for green credit	Accounting for carbon trading
	(1)	(2)	(3)	(4)
$\Delta LNREV$	0.5305*** (49.9474)	0.5304*** (49.9387)	0.5308*** (49.9739)	0.5307*** (49.9627)
$\Delta LNREV \cdot D$	-0.1585** (-2.4038)	-0.1584** (-2.4019)	-0.1591** (-2.4125)	-0.1610** (-2.4403)
$\Delta LNREV \cdot D \cdot LCARBON$	-0.1087*** (-3.0247)	-0.1083*** (-3.0155)	-0.1066*** (-2.9684)	-0.1048*** (-2.9141)
$LCARBON$	0.0035 (0.3315)	0.0033 (0.3165)	0.0004 (0.0377)	-0.0014 (-0.1378)
AIR	-0.0242 (-1.5668)			
$LIMIT$		-0.0208* (-1.7610)		
$GCREDIT$			0.0203 (1.3607)	
$CTRADE$				0.0163 (0.9849)
Other var.	Yes	Yes	Yes	Yes
Constant	-1.3321*** (-10.9212)	-1.3320*** (-10.9240)	-1.3517*** (-11.0867)	-1.3490*** (-11.0658)
N	12,877	12,877	12,877	12,877
FIRM	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Adj. R ²	0.4340	0.4340	0.4339	0.4339

Table 7

Control for various time trend effects

Table 7 presents the regression results for controlling a series of time trends. Column (1) of Table 7 controls city and time trends. Column (2) controls industry and time trends. Column (3) controls two control areas and time trends. Column (4) controls provincial capital cities and time trends. Column (5) controls special economic zones and time trends. Column (6) controls for firms located in the eastern part of China and time trends. For brevity, we do not include the coefficients of control variables. We define the definitions of variables in Appendix A. t-statistics are in the parentheses. ***, **, and * indicate 1%, 5%, and 10%, respectively.

Dep. Var: $\Delta LNSGA$	City and time trends	Industry and time trends	Acid rain-sulfur dioxide and time trends	Provincial capital and time trends	Special economic zone and time trends	Eastern region and time trends
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta LNREV$	0.5306*** (37.0604)	0.5307*** (37.2642)	0.5307*** (37.2372)	0.5306*** (37.1567)	0.5307*** (37.2948)	0.5301*** (36.8789)
$\Delta LNREV \cdot D$	-0.1590** (-2.2655)	-0.1599** (-2.2882)	-0.1598** (-2.2894)	-0.1608** (-2.2896)	-0.1601** (-2.3043)	-0.1580** (-2.2566)
$\Delta LNREV \cdot D \cdot LCARBON$	-0.1062*** (-4.1709)	-0.1066*** (-4.1649)	-0.1066*** (-4.1780)	-0.1072*** (-4.1651)	-0.1063*** (-4.1711)	-0.1064*** (-4.1578)
$LCARBON$	-0.0002 (-0.0152)	0.0000 (0.0021)	0.0001 (0.0105)	-0.0013 (-0.0972)	-0.0002 (-0.0149)	0.0008 (0.0584)
Other var.	Yes	Yes	Yes	Yes	Yes	Yes
$CITY \cdot YEAR$	Yes					
$INDUSTRY \cdot YEAR$		Yes				
$WATCH \cdot YEAR$			Yes			
$CAPITAL \cdot YEAR$				Yes		
$ZONE \cdot YEAR$					Yes	
$EAST \cdot YEAR$						Yes
Constant	-1.3304*** (-9.2052)	-1.3433*** (-8.4439)	-1.3396*** (-8.5496)	-1.3495*** (-8.3429)	-1.3456*** (-8.3689)	-1.1265*** (-7.3853)
N	12, 877	12, 877	12, 877	12, 877	12, 877	12, 877
FIRM	Yes	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.4339	0.4339	0.4339	0.4339	0.4339	0.4341

$SIZE+0.043 * SIZE^2-0.04 * AGE$ |, where $SIZE$ is the natural logarithm of total assets and AGE is the natural logarithm of firm's listing years. The larger the $ABSSA$, the higher the level of financial constraints faced by firms. We follow Brunnermeier & Cohen (2003) to use the number of green patents obtained ($GPATENT$) to measure green innovation. The larger the number of green patents obtained ($GPATENT$), the higher the level of green innovation.

Panel A of Table 8 shows the results of the mediation effect regressions. In the first column, we examine the relation between $LCARBON$ and $ABSSA$. The regression results show that the coefficient of $LCARBON$ is significantly negative at the 1% level, indicating that the low-carbon city policy significantly alleviates corporate financial constraints. In column (2), we test the relation between $LCARBON$ and $GPATENT$. The regression results show that the coefficient of $LCARBON$ is positively significant at the 1% level, indicating that the low-carbon city policy significantly enhances corporate green innovation.

We also present the Bootstrap test results in columns (3) and (4). We conduct 500 random samplings using the Bootstrap method, and the results show that the Bootstrap Z-values of $ABSSA$ and $GPATENT$ are both significant, indicating that financial constraint and green innovation are mediating variables between low-carbon cities and cost stickiness, supporting H2.

Table 8

The mediating role of financial constraint and green innovation

Table 8 shows the regression results of mechanism testing. Panel A shows the regression results of Eq. (3) and Eq. (4). The first column uses $ABSSA$ as the dependent variable. $ABSSA = |-0.737 * SIZE+0.043 * SIZE^2-0.04 * AGE|$, where $SIZE$ is the natural logarithm of total assets and AGE is the natural logarithm of the firm's listing years. The second column uses $GPATENT$ as the dependent variable. $GPATENT$ represents the number of green patents obtained. The third column adds the mediator variable $ABSSA$ to model (1). The fourth column adds the mediator variable $GPATENT$ to model (1). Panel B shows the regression results for the low-carbon policy and loans. Column (1) uses $SHORTLOAN$ as the dependent variable. $SHORTLOAN$ = short-term loans/total assets. Column (2) uses $LONGLOAN$ as the dependent variable. $LONGLOAN$ = long-term loans/total assets. For brevity, we do not include the coefficients of control variables. We define the definitions of variables in Appendix A. t-statistics are in the parentheses. ***, **, and * indicate 1%, 5%, and 10%, respectively.

Panel A				
Dep. Var:	Mediating factors		Bootstrap tests	
	Financial constraints	Green innovation	Financial constraints	Green innovation
	<u>ABSSA</u>	<u>GPATENT</u>	<u>$\Delta LNSGA$</u>	<u>$\Delta LNSGA$</u>
	(1)	(2)	(3)	(4)
$\Delta LNREV$			0.5293*** (49.8205)	0.5310*** (49.9632)
$\Delta LNREV * D$			-0.7681 (-1.0303)	-0.1537** (-2.3211)
$\Delta LNREV * D * LCARBON$			-0.1049*** (-2.9165)	-0.1046*** (-2.9066)
$LCARBON$	-0.0103*** (-4.8410)	2.8937*** (3.5909)	0.0015 (0.1431)	-0.0002 (-0.0241)
$\Delta LNREV * D * ABSSA$			0.1911 (0.8267)	
$ABSSA$			0.1675*** (3.4777)	
$\Delta LNREV * D * GPATENT$				-0.0017 (-1.1173)
$GPATENT$				0.0001 (0.4652)
Other var.	Yes	Yes	Yes	Yes
Constant	2.5251*** (106.5417)	-37.7702*** (-4.2077)	-1.7913*** (-10.1022)	-1.3377*** (-10.9724)
N	12,877	12,862	12,877	12,862
FIRM	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes
Adj. R ²	0.8131	0.7221	0.4344	0.4339
Bootstrap Z			19.98***	1.85*
Panel B				
Dep. Var	<u>SHORTLOAN</u>		<u>LONGLOAN</u>	
	(1)		(2)	
$LCARBON$	-0.0003 (-0.1024)		0.0074*** (2.9611)	
Other var.	Yes		Yes	
Constant	0.2800*** (8.6246)		-0.3151*** (-10.8954)	
N	11,144		8350	
FIRM	Yes		Yes	
YEAR	Yes		Yes	
Adj. R ²	0.7601		0.5541	

In addition, in Panel B of Table 8 we conduct a specific analysis on the path of financial constraints. Following Lin (2011), we use short-term loans/total assets to represent short-term loans (*SHORTLOAN*), and long-term loans/total assets to represent long-term loans (*LONGLOAN*). We take short-term loans and long-term loans as the dependent variables, and *LCARBON* as the explanatory variable. The regression results are shown in Panel B of Table 8. In Column (1) of Panel B, the coefficient of *SHORTLOAN* is insignificant. In Column (2) of Panel B, the coefficient of *LONGLOAN* is significantly positive at the 1% level, indicating that the mechanism of low-carbon city policy to alleviate financial constraints is mainly reflected in increasing the level of long-term loans for firms.

4.5. Moderating analysis

4.5.1. Traditional manufacturing industries vs. modern manufacturing industries

Traditional manufacturing industries have high energy consumption and environmental pollution. Under the influence of low-carbon city policy, traditional manufacturing industries are more motivated to invest in green innovation, leading to an increase in cost stickiness. Therefore, we divide the sample into traditional manufacturing and modern manufacturing industries. The modern manufacturing industry mainly includes automobile manufacturing, railway, shipbuilding, aerospace and other transportation equipment manufacturing, electrical machinery and equipment manufacturing, computer, communication and other electronic equipment manufacturing, instrument manufacturing, and pharmaceutical manufacturing enterprises. The results in columns (1) and (2) of Table 9 show that only the coefficient of $\Delta LNREV \cdot D \cdot LCARBON$ is negatively significant for the firms in the traditional manufacturing industries, which is consistent with our conjecture.

4.5.2. State-ownership

State-owned firms (SOEs), due to their state backing, have lower financial constraints than non-SOEs. When a SOE (non-SOE) is located in a low-carbon city, it has more (fewer) resources to compile with the environment-related initiatives. Therefore, non-SOEs can better leverage the low-carbon city policy to obtain more funding. Therefore, cost stickiness is more salient for non-SOEs. The results in columns (3) and (4) of Table 9 are consistent with the conjecture.

4.5.3. Technology level of a firm

Green innovation typically requires technology. Thus, a high technology firm prefers to engage in green innovation to address environmental policy requirement than that of a non-high technology firm. Such a preference of using green innovation increases the cost stickiness for a high technology firm more than that of a non-high technology firm. Specifically, when a firm has above (below) median intangible assets in its industry, it belongs to a high-tech (low-tech) industry. The results in columns (5) and (6) of Table 9 show that the coefficient of $\Delta LNREV \cdot D \cdot LCARBON$ continues to be negatively significant at the 1% level for the high technology firm subsample while the coefficient is insignificant for the opposite subsample.

4.6. Economic consequence analysis

We examine the economic consequence of firms that exhibit an increase in cost stickiness. Following Weiss (2010), we use quarterly

Table 9

Moderating analysis

Table 9 presents the results of heterogeneity analysis. The sample are divided into traditional manufacturing and modern manufacturing industries in columns (1) and (2). According to the nature of property rights, the samples are divided into state-owned and non-state-owned firms in columns (3) and (4). Columns (5) and (6) divide the sample into high-tech and low-tech firms based on the median of intangible assets. For brevity, we do not include the coefficients of control variables. We define the definitions of variables in Appendix A. t-statistics are in the parentheses. ***, **, and * indicate 1%, 5%, and 10%, respectively.

Dep. Var: $\Delta LNSGA$	Industry		State-ownership		Technology level	
	Traditional manufacturing industry	Modern manufacturing industry	Yes	No	Low	High
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta LNREV$	0.5023*** (34.4865)	0.5595*** (34.7387)	0.5940*** (17.0948)	0.5286*** (33.6123)	0.4674*** (19.3695)	0.5638*** (25.3824)
$\Delta LNREV \cdot D$	-0.1555* (-1.7817)	-0.2225** (-2.1532)	-0.0733 (-0.2070)	-0.1476** (-2.0606)	-0.0685 (-0.7612)	-0.0952 (-0.7334)
$\Delta LNREV \cdot D \cdot LCARBON$	-0.1506*** (-3.2107)	-0.0795 (-1.3411)	-0.0462 (-0.4194)	-0.1301*** (-4.1215)	-0.0486 (-1.0173)	-0.1479*** (-3.2240)
<i>LCARBON</i>	0.0009 (0.0725)	0.0028 (0.1430)	0.0211 (0.7325)	-0.0065 (-0.5546)	0.0004 (0.0195)	-0.0082 (-0.6098)
<i>Other var.</i>	Yes	Yes	Yes	Yes	Yes	Yes
Constant	-1.3924*** (-7.9397)	-1.6918*** (-8.8264)	-0.7658** (-2.3699)	-1.3825*** (-8.4617)	-1.8030*** (-5.0504)	-1.0997*** (-3.3279)
N	7334	5543	1, 509	11, 368	6, 437	6, 437
FIRM	Yes	Yes	Yes	Yes	Yes	Yes
YEAR	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.4086	0.4722	0.4182	0.4400	0.4294	0.4628

financial data to calculate a firm's annual cost stickiness and take absolute value of the metric (denoted as *ABSSTICKY*). A large value of *ABSSTICKY* suggests a large magnitude of cost stickiness. Then, we calculate a firm's total factor productivity using alternative metrics of total factor productivity (*TFP_OP*). Specifically, we follow [Olley & Pakes \(1996\)](#) to estimate a firm's Cobb-Douglas production function as follow:

$$\ln Y_{i,t} = \eta_0 + \eta_1 \ln K_{i,t} + \eta_2 \ln L_{i,t} + \xi_{i,t} \quad (5)$$

Where *Y* is a firm's output (defined as operating income/10,000), *K* is capital (defined as net fixed assets/10,000), and *L* is labor (as the total number of employee). To reduce the simultaneity bias and sample selection bias in the ordinary least squares estimation of Eq. (5), we use the semi-parametric three-step estimation method of [Olley & Pakes \(1996\)](#). The residual from Eq. (5) is total factor productivity of a firm (denoted as *TFP_OP*).

Then, we estimate the following regression equation:

$$TFP_OP_{i,t} = \lambda_0 + \lambda_1 ABSSTICKY_{i,t} + CONTROL + YEAR + FIRM + \mu_{i,t} \quad (6)$$

$$TFP_OP_{i,t} = \pi_0 + \pi_1 ABSSTICKY_{i,t} + \pi_2 LCARBON_{i,t} * ABSSTICKY_{i,t} + \pi_3 LCARBON_{i,t} + CONTROL + YEAR + FIRM + \nu_{i,t} \quad (7)$$

The set of control variables is the same as those in Eq. (1). We present the findings in [Table 10](#). Column (1) of [Table 10](#) shows that the coefficients of *ABSSTICKY* are consistently negative and significant at the 5% level, suggesting that when a firm has high cost stickiness, it generally has low TFP. The findings echo [Costa & Habib \(2023\)](#) who report cost stickiness destroys firm value. That is when a firm has capacity slack (due to cost stickiness), it has adverse effect on operation efficiency. However, Column (2) of [Table 10](#) shows that the coefficients of *LCARBON*ABSSTICKY* are positively significant at the 1% level. As shown in Eq. (7), when *LCARBON* equals to 1, the coefficient before *ABSSTICKY* is $(\pi_1 + \pi_2)$. Therefore, when the firm is in a low-carbon city, the absolute value of the coefficient before *ABSSTICKY* in Column (2) $(|\pi_1 + \pi_2| = |-0.0139 + 0.0132| = 0.0007)$ is significantly smaller than that in Column (2) $(|\lambda_1| = |-0.058| = 0.058)$. Thus, the low-carbon city policy can significantly absorb the adverse effects of cost stickiness, provide a resource base for enterprise development, and thus improve total factor productivity.

5. Conclusions

We examine the impact of a low-carbon city policy on corporate cost stickiness of firms located in such a city. Drawing from the staggered events of designating selected Chinese cities as low-carbon cities, we apply a DID research design to conduct our analysis.

Our findings suggest that cost stickiness increases for firms in a city with a low-carbon policy in a sample of Chinese firms. The findings are robust to an entropy balance method, alternative samples, placebo tests, after accounted for other environmental regulations, and after control for various time trend effects. The mechanism analysis suggests that easing financial constraints and engaging in green investments are the underlying logic behind the impact of low-carbon city policy on corporate cost stickiness. Additional analysis shows that the effect is more salient for firms in the traditional manufacturing industries, which are state-owned, or belong to high technology industries. The economic consequences find that although the low-carbon city policy will increase cost stickiness, with the implementation of the policy, it will absorb the adverse effects of cost stickiness, provide resources and foundation for enterprise development, and thus improve the total factor productivity of enterprises.

Our results have several policy implications. First, the implementation of low-carbon city policy allows firms in the city to ease their financial constraint and drive them to engage in more green innovation. Thus, instead of unselectively giving out environmental

Table 10

Economic consequence analysis

[Table 10](#) presents the results for the economic consequence of firms having cost stickiness due to low-carbon city designation. The dependent variables are total factor productivity (*TFP_OP*). For brevity, we do not include the coefficients of control variables. We define the definitions of variables in [Appendix A](#). t-statistics are in the parentheses. ***, **, and * indicate 1%, 5%, and 10%, respectively.

Dep. Var: <i>TFP_OP</i>	(1)	(2)
<i>ABSSTICKY</i>	-0.0058** (-2.0853)	-0.0139*** (-3.1562)
<i>LOWCARBON*ABSSTICKY</i>		0.0132** (2.3777)
<i>LOWCARBON</i>		0.0010 (0.0721)
Other var.	Yes	Yes
Constant	-1.2868*** (-8.0284)	-1.2938*** (-8.0305)
N	11,432	11,432
FIRM	Yes	Yes
YEAR	Yes	Yes
Adj. R ²	0.9119	0.9119

subsidies or reduce taxes for environmental projects, it is helpful to designate a number of low-carbon cities and focus the government resources on firms in these cities. While an increase in cost stickiness generally means a lower productivity, we document that the benefits of operating in a low-carbon city offset the adverse effect of productivity. Given no effect on productivity and the benefits of low-carbon city policy, China offers a good lesson for other emerging markets to implement similar policy.

There are some limitations. We only use the manufacturing industry samples instead of industry wide samples, which might limit the generalizability of the findings to other sectors. Future research could focus on exploring the relation between low-carbon city policy and cost stickiness in industry-wide samples. Besides, we could also do some comparative researches involving similar policies in different country contexts, which could provide insights into how cultural, economic, and regulatory differences influence the effectiveness of such environmental policies.

Data availability

Data will be made available on request.

Appendix A. Variable definitions

	Variables	Definitions	Data source
Dependent variables	$\Delta LNSGA$	The change in natural logarithm of sales, general, and administrative expense from $t-1$ to t for firm i .	CSMAR
	SGA	The sales, general, and administrative expense at t for firm i (million RMB)	CSMAR
	TFP_{OP}	We calculate total factor productivity (TFP) using Olley & Pakes (1996) method	CSMAR
Explanatory variables	$\Delta LNREV$	The change in natural logarithm of revenue from $t-1$ to t for firm i .	CSMAR
	REV	The revenue at t for firm i (million RMB)	CSMAR
	D	A dummy variable with a value of 1 if REV at t is less than REV at $t-1$ and 0 otherwise.	CSMAR
	$LCARBON$	A dummy variable with a value of 1 if a firm is located in a low-carbon policy city at t and 0 otherwise.	CSMAR
	$ABSSTICKY$	Following Weisse (2010) , we use quarterly financial data to calculate a firm's annual cost stickiness and take absolute value of the metric	CSMAR
Control variables	MSH	The ratio of shareholding of directors, supervisors, and senior executives to outstanding shares	CSMAR
	AGE	The natural logarithm of the number of listing years	CSMAR
	SHB	The ratio of the sum of second to fifth largest shareholding to the top shareholder's	CSMAR
	$SIZE$	The natural logarithm of the total assets	CSMAR
	ROA	The ratio of net income to total assets	CSMAR
	LEV	The ratio of total liabilities to total assets	CSMAR
	SOE	A dummy variable with a value of 1 if a firm is state-owned and 0 otherwise	CSMAR
	$DECREASE$	A dummy variable with a value of 1 if a firm has two consecutive years decrease in revenue and 0 otherwise	CSMAR
	$GDPGROWTH$	The growth rate of GDP at a city where a firm is located	CSMAR
	$AINTENSITY$	The ratio of total revenue to total assets	CSMAR
	$EINTENSITY$	The ratio of total revenue to the total number of employees (in millions RMB)	CSMAR
Moderating variables	$ABSSA$	$ABSSA = -0.737 * SIZE + 0.043 * SIZE^2 - 0.04 * AGE $, where $SIZE$ is the natural logarithm of total assets and AGE is the natural logarithm of firm's listing years.	CSMAR
	$GPATENT$	The number of green patents held in a year	CNRDS
	$INTANGIBLE$	The value of intangible assets in a year	CSMAR

Appendix B. Entropy balancing results

[Appendix B](#) shows the sample size, mean, variance, skewness, and differences in mean, variance, and skewness of the covariates between the experimental and control groups after entropy balance matching (Difference is obtained by subtracting the mean, variance, and skewness of the experimental and control groups, respectively).

	Treatment group				Control group (weighted)				Difference		
	N	Mean	Variance	Skewness	N	Mean	Variance	Skewness	Mean	Variance	Skewness
MSH	7816	21.950	441.100	0.538	5061	21.950	441.100	0.538	0.000	0.000	0.000
AGE	7816	1.788	0.271	-0.156	5061	1.788	0.271	-0.156	0.000	0.000	0.000
SHB	7816	0.862	0.404	1.130	5061	0.862	0.404	1.130	0.000	0.000	0.000
$SIZE$	7816	21.860	1.048	0.788	5061	21.860	1.048	0.788	0.000	0.000	0.000
ROA	7816	0.040	0.004	-1.396	5061	0.040	0.004	-1.396	0.000	0.000	0.000
LEV	7816	0.365	0.031	0.221	5061	0.365	0.031	0.221	0.000	0.000	0.000
SOE	7816	0.122	0.107	2.315	5061	0.122	0.107	2.315	0.000	0.000	0.000
$DECREASE$	7816	0.088	0.080	2.908	5061	0.088	0.080	2.908	0.000	0.000	0.000
$GDPGROWTH$	7816	5.979	4.313	-0.681	5061	5.979	4.313	-0.680	0.000	0.000	0.000
$AINTENSITY$	7816	2.262	1.644	1.873	5061	2.262	1.644	1.873	0.000	0.000	0.000
$EINTENSITY$	7816	1.376	0.710	1.271	5061	1.376	0.710	1.271	0.000	0.000	0.000

References

- Anderson, M. C., Banker, R. D., & Janakiraman, S. N. (2003). Are selling, general, and administrative costs "sticky"? *Journal of Accounting Research*, 41(1), 47–63.
- Bai, X., Han, J., Ma, Y., & Zhang, W. (2022). ESG performance, institutional investors' preference and financing constraints: Empirical evidence from China. *Borsa Istanbul Review*, 22, S157–S168.
- Banker, R. D., & Byzalov, D. (2014). Asymmetric cost behavior. *Journal of Management Accounting Research*, 26(2), 43–79.
- Banker, R. D., Byzalov, D., & Chen, L. T. (2013). Employment protection legislation, adjustment costs and cross-country differences in cost behavior. *Journal of Accounting and Economics*, 55(1), 111–127.
- Banker, R. D., Byzalov, D., & Plehn-Dujowich, J. M. (2014). Demand uncertainty and cost behavior. *The Accounting Review*, 89(3), 839–865.
- Baron, R. M., & Kenny, D. A. (1986). The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, 51(6), 1173–1182.
- Belina, H., Suryekar, K., & Weismann, M. (2019). On the medical loss ratio (MLR) and sticky selling general and administrative costs: Evidence from health insurers. *Journal of Accounting and Public Policy*, 38(1), 53–61.
- Brunnermeier, S. B., & Cohen, M. A. (2003). Determinants of environmental innovation in US manufacturing industries. *Journal of Environmental Economics and Management*, 45(2), 278–293.
- Cai, X., Huang, Y., Ying, S. X., & Chen, H. (2023). Can climate policy promote high-quality development of enterprises? Evidence from China. *Frontiers in Environmental Science*, 11.
- Cao, J., Li, W., & Hasan, I. (2023). The impact of lowering carbon emissions on corporate labour investment: A quasi-natural experiment. *Energy Economics*, 121, Article 106653.
- Chang, H., Dai, X., Lohwasser, E., & Qiu, Y. (2022). Organized labor effects on SG&A cost behavior. *Contemporary Accounting Research*, 39(1), 404–427.
- Chen, H., Guo, W., Feng, X., Wei, W., Liu, H., Feng, Y., & Gong, W. (2021). The impact of low-carbon city pilot policy on the total factor productivity of listed enterprises in China. *Resources, Conservation and Recycling*, 169, Article 105457.
- Chen, J. V., Kama, I., & Leavy, R. (2019). A contextual analysis of the impact of managerial expectations on asymmetric cost behavior. *Review of Accounting Studies*, 24(2), 665–693.
- Chen, C. X., Lu, H., & Sougiannis, T. (2012). The agency problem, corporate governance, and the asymmetrical behavior of selling, general, and administrative costs. *Contemporary Accounting Research*, 29(1), 252–282.
- Costa, M. D., & Habib, A. (2023). Cost stickiness and firm value. *Journal of Management Control*, 34, 235–273.
- Dai, J., Hu, N., Huang, R., & Yan, Y. (2023). How does credit risk affect cost management strategies? Evidence on the initiation of credit default swap and sticky cost behavior. *Journal of Corporate Finance*, 80, Article 102401.
- Duan, M., & Xu, R. (2023). Does low-carbon city pilot affect the enterprise competitiveness in China? Based on a staggered difference-in-difference model. *Environmental Science and Pollution Research*, 30(13), 37233–37247.
- Ferri, F., & Li, N. (2020). Does option-based compensation affect payout policy? Evidence from FAS 123R. *Journal of Financial and Quantitative Analysis*, 55(1), 291–329.
- Gao, H., Hsu, P., Li, K., & Zhang, J. (2020). The real effect of smoking bans: Evidence from corporate innovation. *Journal of Financial and Quantitative Analysis*, 55(2), 387–427.
- Hadlock, C. J., & Pierce, J. R. (2010). New evidence on measuring financial constraints moving beyond the KZ index. *Review of Financial Studies*, 23(5), 1909–1940.
- Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, 20(1), 25–46.
- Holzhaecker, M., Krishnan, R., & Mahlendorf, M. D. (2015). The impact of changes in regulation on cost behavior. *Contemporary Accounting Research*, 32(2), 534–566.
- Huang, S., Roychowdhury, S., & Sletten, E. (2019). Does litigation deter or encourage real earnings management? *The Accounting Review*, 95(3), 251–278.
- Huang, J., Zhao, J., & Cao, J. (2021). Environmental regulation and corporate R&D investment—evidence from a quasi-natural experiment. *International Review of Economics & Finance*, 72, 154–174.
- Kama, I., & Weiss, D. (2013). Do earnings targets and managerial incentives affect sticky costs? *Journal of Accounting Research*, 51(1), 201–224.
- Li, J., Lian, G., & Xu, A. (2023). How do ESG affect the spillover of green innovation among peer firms? Mechanism discussion and performance study. *Journal of Business Research*, 158, Article 113648.
- Li, Z., Liao, G., Wang, Z., & Huang, Z. (2018). Green loan and subsidy for promoting clean production innovation. *Journal of Cleaner Production*, 187, 421–431.
- Lin, H. (2011). Foreign bank entry and firms' access to bank credit: Evidence from China. *Journal of Banking & Finance*, 35(4), 1000–1010.
- Lin, Y., & Fang, P. (2024). ESG performance and corporate cost stickiness. Available at: SSRN 4620686.
- Olley, G. S., & Pakes, A. (1996). The dynamics of productivity in the telecommunications equipment industry. *Econometrica*, 64, 1263–1297.
- Pei, F., & Wang, P. (2022). The impact of the low-carbon city pilot policy on green innovation in firms. *Frontiers in Environmental Science*, 10, 1765.
- Porter, M. E., & Linde, C. V. D. (1995). Toward a new conception of the environment-competitiveness relationship. *The Journal of Economic Perspectives*, 9(4), 97–118.
- Qu, F., Xu, L., & He, C. (2023). Leverage effect or crowding out effect? Evidence from low-carbon city pilot and energy technology innovation in China. *Sustainable Cities and Society*, 91, Article 104423.
- Wang, T., Song, Z., Zhou, J., Sun, H., & Liu, F. (2022). Low-carbon transition and green innovation: Evidence from pilot cities in China. *Sustainability*, 14(12), 7264.
- Weiss, D. (2010). Cost behavior and analysts' earnings forecasts. *The Accounting Review*, 85(4), 1441–1471.
- Wu, H., Wang, L., Ding, T., & Peng, F. (2024). Does urban low-carbon governance affect firms' business decisions for employing capital and labor. *Journal of Environmental Planning and Management*, 67(10), 2257–2280.
- Xu, H., Chan, K. C., Na, C., & Fang, Q. (2023). The bright side of the internal labor market: Evidence from the labor cost stickiness of firms affiliated with privately owned business groups in China. *Journal of Corporate Finance*, 78, Article 102356.
- Yang, S., Jahanger, A., & Hossain, M. R. (2023). How effective has the low-carbon city pilot policy been as an environmental intervention in curbing pollution? Evidence from Chinese industrial enterprises. *Energy Economics*, 118, Article 106523.
- Yang, Z., & Xu, Y. (2023). Do different types of carbon mitigation regulations have heterogeneous effects on innovation quality? *Environmental Science and Pollution Research*, 30(15), 43168–43182.
- Zeng, S., Jin, G., Tan, K., & Liu, X. (2023). Can low-carbon city construction reduce carbon intensity? Empirical evidence from low-carbon city pilot policy in China. *Journal of Environmental Management*, 332, Article 117363.
- Zhang, P., Zhou, D., & Guo, J. (2023). Policy complementary or policy crowding-out? Effects of cross-instrumental policy mix on green innovation in China. *Technological Forecasting and Social Change*, 192, Article 122530.
- Zhu, Y., & Sun, C. (2022). Carbon reduction, pollution intensity, and firms' ratios of value added in exports: Evidence from China's low-carbon pilot policy. *Sustainability*, 14(19), Article 12687.
- Zhuo, W. (2023). Environmental regulation and corporate sustainability: Evidence from green innovation. *Corporate Social Responsibility and Environmental Management*, 30(4), 1723–1737.